MSCI World Factor Strategy

Portfolio backtest · Research report · 14 October 2026 USD results · 3 October 2014-28 August 2026
AI-assisted analysis and writing; tools, scope and responsibility are disclosed in the final section.

Abstract

This study evaluates a shared MSCI World portfolio with Core, Momentum, Quality and Value weights of 60/15/10/15 and a 1.25x leverage target. Over 3 October 2014-28 August 2026, its CAGR is **13.24%**, compared with **12.82%** for the MSCI World/Core investment at the same target leverage. Both include modelled trading and financing costs; the portfolio result excludes custody charges, the added fund fee and outside-fund business costs. At the end of 2025, matched-target wealth was nearly equal; the positive terminal wealth advantage emerged in the observed 2026 segment. The full-sample comparison is endpoint-sensitive. Borrowing raises return and risk relative to the unlevered portfolio. The estimated alpha against unlevered Core is -0.08% annually, with an interval including zero. The concept offers shared factor and leverage access; the historical evidence supports neither reliable future outperformance nor a proven commercial advantage.

1. Research Question

How does a shared portfolio anchored in the MSCI World, with momentum, quality and value sleeves, compare with a broad-market investment in growth, risk and cost? Does borrowing add return after investment costs, and what additional risks accompany it?

The intended value is a rules-based factor/leverage alternative for investors for whom direct borrowing is unavailable or relatively expensive. Its usefulness need not depend on persistent benchmark outperformance. Actual borrowing access and terms must still be established.

The main strategy uses 1.25x target leverage. Unlevered results provide a control for the contribution and risk of borrowing. A favourable joint historical comparison requires higher CAGR, no higher volatility and no deeper maximum drawdown. This criterion is distinct from statistical alpha or a reliable forecast of future outperformance.

2. Portfolio Design and Benchmark

2.1 Allocation and trading rules

Table 1. Target allocation · Target weights: backtest configuration

Sleeve	Target	Economic role
Core	60.00%	Broad developed-market anchor
Momentum	15.00%	Relative-performance selection
Quality	10.00%	Financial-quality selection
Value	15.00%	Valuation selection

The Core provides broad developed-market equity exposure; the factor sleeves apply different selection and weighting characteristics within the MSCI World framework. They remain equity investments and can decline together. The 60% Core keeps broad-market exposure central, while the remaining 40% diversifies the proposed factor tilts. The 1.25x target adds moderate borrowing rather than adopting a daily 2x product. These are design choices, not estimates of optimal future weights.

Weights are measured against gross invested assets. Each sleeve may drift by five percentage points above or below its target: Core 55-65%, Momentum 10-20%, Quality 5-15% and Value 10-20%. Equality at a boundary does not

trigger trading. A strict breach signals a full sleeve reset at the next common NAV observation. A sleeve-only reset preserves debt; a leverage-only adjustment scales the existing mixture proportionally. A simultaneous signal resets both.

Leverage is managed separately at 1.25x with a ±0.10 band. This is an adjustment rule, not a hard cap on realised leverage. An account begins with USD 7.00 million of committed equity, corresponding to USD 8.75 million of gross exposure and USD 1.75 million of debt before opening charges. Portfolio and cost settings: backtest configuration.

The proposed opening equity represents seven founders contributing USD one million each. The backtest treats their investment as one aggregate account; it does not model separate founder transactions.

Five-percentage-point bands were the author's initial band-width proposal. Tighter alternatives were explored before returning to that width for simplicity and greater permitted factor drift. This chronology does not make the final specification independent of the historical evidence. The separated leverage mechanism must also be distinguished from coupled resets in some exploratory controls. A capped-relative rule, using the smaller of five percentage points and twenty percent of each sleeve target, is retained as a sensitivity in Appendix A.

2.2 Why MSCI World is the benchmark

The MSCI World is the primary market benchmark because it supplies the common developed-market framework for the Core and factor sleeves. A broad-market investment is the natural alternative against which to evaluate the combined allocation and management rules. The 60% Core allocation reinforces this connection; the common framework is the principal justification.

The backtest implements this benchmark through the **iShares Core MSCI World UCITS ETF**, an investable proxy whose NAV includes embedded fund expenses. It is not the bare index return. Results compare both unlevered investments and investments at the same 1.25x target. Equal target exposure does not establish equal realised leverage, beta, costs or investment universes.

Dimensional is a secondary systematic-investment alternative. Amundi is a short-period real leveraged-product comparison. Neither replaces the MSCI World/Core benchmark.

3. Data and Method

3.1 Inputs and common calendar

Table 2. Main analytical inputs · NAV and funding inputs: backtest_sources.json

Source	Period	Use
BlackRock/iShares accumulating ETF NAV exports	Common sample: 3 October 2014-28 August 2026	Core, Momentum, Quality and Value investment returns
Author-supplied Bloomberg Dimensional NAV hardcopy	Same common sample	Secondary systematic-fund comparison
New York Fed indicative reference series and official SOFR	Relevant financing dates	Reference rate for borrowed exposure
Saved Amundi NAV workbook	Separate overlap: 30 September 2025-31 August 2026	Real daily 2x product comparison

The main common calendar contains **2929 levels and 2928 adjacent return intervals**. Investment levels are intersected before calculating returns; missing investment NAVs are not filled. The short Amundi experiment uses its own overlap and extends to August month-end. Reference rates are carried across nonpublication calendar days. The early indicative series is distinct from later official SOFR; both are financing proxies rather than the actual account's borrowing rate. Common NAV calendar: backtest_workflow.prepare_inputs.

Accumulating NAVs include embedded fund expenses and reinvestment treatment. Deducting TER again would double-count that layer. NAV is not an executable bid or ask. Investor trading costs are therefore specified separately. Published rounded issuer returns provide a secondary check for 40 calendar-year observations; the largest central-value difference is 0.07 pp and remains compatible with the retained rounding assumptions. This is not a second complete daily-price feed. Issuer return check: backtest_issuer_returns.run_issuer_returns.

The Dimensional comparison uses the Global Core Equity Fund, USD Accumulation Shares (ISIN IE00B2PC0153), with NAV observations from the supplied Bloomberg workbook. Its investment mandate differs from MSCI World and allows some emerging-market investments. That permission does not establish the fund's actual emerging-market allocation throughout the sample. Fund identities and sources are listed after the conclusion.

3.2 Accounting and costs

Gross invested assets minus debt equals investor equity. At an observation, positions are marked and financing is accrued; prior signals are executed before new signals are evaluated. Trading charges reduce equity. No subscriptions or withdrawals enter the investment backtest. Trading, debt and fees: historical.simulate.

Table 3. Investment cost and measurement assumptions · Trading and borrowing costs: backtest configuration

Baseline parameter	Selected value
Financing markup above reference	3 percentage points (300 bp)
Reference day count	ACT/360
Platform fee per traded leg	USD 0.85
Investor-side duty	0.15%
Ordinary one-way spread	5 bp
USD per CHF	1.2368

The borrowing charge is the reference rate **plus 3 percentage points (300 bp)**. Each traded leg also incurs a stepped broker commission, ranging from CHF 3 to CHF 190 according to the configured trade-notional brackets. The commission is converted into USD at the fixed rate below; the USD 0.85 platform charge is additional. Broker commission: config.SwissquoteStandardFeeSchedule specifies the exact brackets used in this current-cost scenario.

Funding accrues from the previous observation inclusive to the current observation exclusive, on ACT/360, and is capitalised at NAV dates. Current-cost scenarios apply the stated tariff and commission conversion throughout history rather than reconstructing historical invoices. Fund administration charges, custody sensitivities, investor taxes and manager business budgets are outside the baseline investment account. Trading, debt and fees: historical.simulate.

The fixed commission conversion is **1 CHF = USD 1.2368**, derived from the recorded ECB reference rates for **31 August 2026**: 1.1596 USD per EUR divided by 0.9376 CHF per EUR. Only the CHF-denominated broker commission is converted; investment returns and debt already use USD. Full precision is retained in the calculation. This endpoint reference rate is a modelling assumption, not a historical series of broker execution rates. Portfolio and cost settings: backtest configuration.

CAGR uses committed opening capital and elapsed calendar years. Opening trading charges are included once and folded into the first investor-return interval. Drawdown uses a high-water mark no lower than committed opening capital. Volatility uses 252-observation annualisation. Appendix C explains the calendar, annualisation and financing conventions. Jensen alpha is an annualised excess-return regression intercept against unlevered Core, not the CAGR difference; exploratory intervals use five-lag HAC errors. Return and risk metrics: historical.metrics.

Separate implementations check account identities and selected daily, event and outcome results under shared inputs and conventions. These checks address implementation errors; they do not establish market-price executability, commercial feasibility or expert peer review.

4. Historical Results

4.1 Central MSCI World comparison

Table 4. Unlevered control over the main sample · CAGR, volatility and drawdown: historical.metrics

Investment / rule	CAGR	Volatility	Drawdown	Sharpe	Ending equity, USD m
Absolute / separate leverage	11.87%	14.96%	-33.56%	0.702	26.59
MSCI World Core	11.52%	15.18%	-34.07%	0.672	25.62

Table 5. Main portfolio and Core at a 1.25x target · CAGR, volatility and drawdown: historical.metrics

Investment / rule	CAGR	Volatility	Drawdown	Sharpe	Ending equity, USD m
Absolute / separate leverage	13.24%	18.59%	-39.70%	0.665	30.76
MSCI World Core	12.82%	18.94%	-40.36%	0.636	29.43

These account returns include embedded product expenses, investor trading charges and financing. They exclude custody charges, the additional fund fee and the outside-fund operating budget analysed in the companion study; the separate custody sensitivity is identified in Appendix C. The main strategy meets the stated joint growth/risk criterion against Core over the full sample. Its CAGR advantage is 0.35 pp without borrowing and 0.42 pp at matched target leverage. These are historical account comparisons, not a causal estimate of the return from each factor. Return and risk metrics: historical.metrics.

Borrowing increases portfolio CAGR from 11.87% to 13.24%, but also raises volatility from 14.96% to 18.59% and deepens maximum drawdown from -33.56% to -39.70%. The additional growth comes with additional total risk. Return and risk metrics: historical.metrics.

Table 6. Main strategy and the unlevered retail alternative · Existing return and risk measures: reader_diagnostics.summaries

Investment	Target	CAGR	Volatility	Sharpe
Factor portfolio	1.25x	13.24%	18.59%	0.665
MSCI World Core	1x	11.52%	15.18%	0.672

For an investor without convenient borrowing access, unlevered Core is a relevant practical alternative. The main portfolio delivered higher historical CAGR, with higher volatility and a slightly lower Sharpe ratio, before the added fund fee. Its value proposition is access to a chosen factor/leverage exposure; these figures do not establish better risk-adjusted performance than a simple Core holding. The matched-target comparison remains necessary to distinguish this question from the effects of borrowing.

Table 7. End-date sensitivity of the continuing investment paths · Relative wealth: reader_diagnostics.summaries

Last common NAV	Relative wealth vs Core 1.25x
2022-12-30	+1.21%
2023-12-29	-1.17%
2024-12-31	-2.19%
2025-12-31	-0.07%
2026-08-28	+4.51%

Relative wealth is 100 x (portfolio equity / Core equity - 1), with both paths at a 1.25x target and unchanged opening capital. These are checkpoints of the continuing accounts, not independently restarted backtests or annual return gaps. The portfolio recovered most of its 2024 shortfall during 2025 and was nearly level with Core at the end of 2025. Its positive terminal wealth advantage emerged in the observed 2026 segment. The headline result is therefore sensitive to the chosen endpoint and does not demonstrate a persistent advantage throughout the sample.

After the opening investment, the main path contains only **two rule-triggered interventions**: a sleeve reset on **3 September 2020** and a leverage adjustment on **9 April 2021**. The count includes leverage adjustments; each intervention can comprise several individual purchases or sales. Most of this historical path therefore consists of held positions with permitted weight and leverage drift. The thresholds still govern when trading is avoided, and few interventions do not remove retrospective selection risk. The event ledger records both actions.

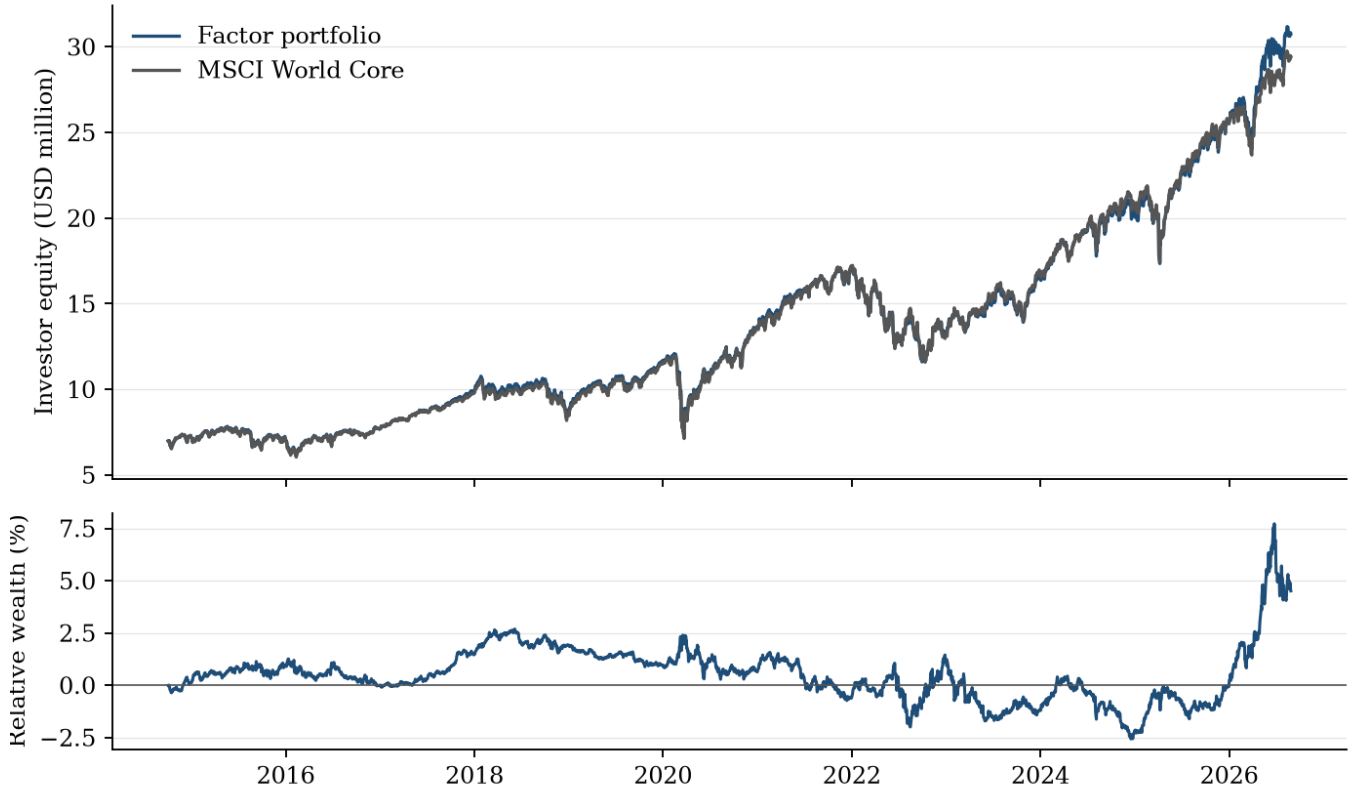


Figure 1. Main portfolio and Core at the same 1.25x target, after modelled trading and financing. The lower panel shows their relative ending wealth at each date: 100 x (portfolio equity / Core equity - 1). This is not an annual return or regression alpha. The capped-relative comparison remains in Appendix A. Reader charts from recorded evidence: reader_figures.render

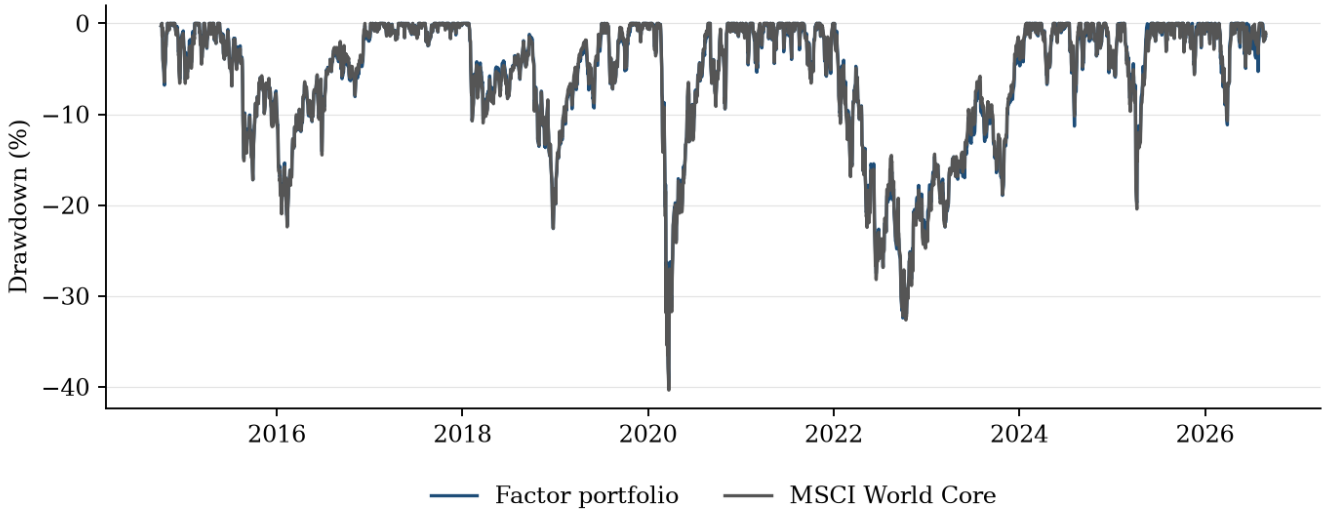


Figure 2. Main portfolio and Core drawdowns from their committed-capital high-water marks. Opening charges are included; intraday lender exposure is outside this observation calendar. Reader charts from recorded evidence: reader_figures.render

Table 8. Realised leverage on the common NAV calendar · Leverage summaries: reader_diagnostics.summaries

Observed portfolio leverage	Value	Date / scope
Observation mean	1.232x	All observations
Minimum	1.150x	2021-04-08
Maximum	1.345x	2022-10-12
Final observation	1.175x	2026-08-28

The observation mean is close to the target and to the Core comparator's **1.233x**. Leverage varies because market moves change equity and financing accrues into debt. The 1.25x label is an actively monitored target rather than a constant daily exposure or only an opening value. Similar average leverage does not imply equal daily exposure, beta or risk. The observed maximum is a common-NAV measure, not an intraday maximum or a verified distance to a lender's margin-call threshold.

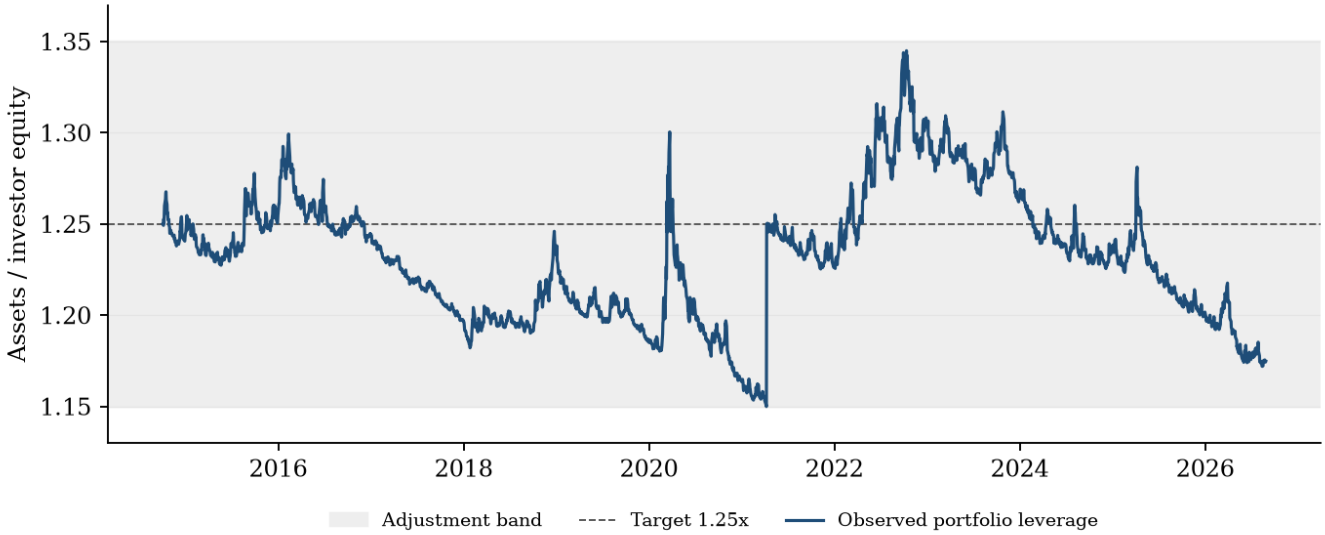


Figure 3. Observed portfolio leverage, target and adjustment band. Strict breaches are executed at the next common NAV; the band is not a hard realised-exposure cap. The average uses the same observation-based definition as the reported investment metrics. Recorded leverage plot: reader_figures.render

4.2 Brief comparison with Dimensional

Table 9. Main strategy and Dimensional at the 1.25x target · CAGR, volatility and drawdown: historical.metrics

Investment / rule	CAGR	Volatility	Drawdown	Sharpe	Ending equity, USD m
Absolute / separate leverage	13.24%	18.59%	-39.70%	0.665	30.76
Dimensional	11.32%	20.92%	-43.55%	0.529	25.08

The main portfolio's CAGR advantage over Dimensional is 1.93 pp, with lower observed volatility and a shallower maximum drawdown. Dimensional is an existing systematic fund alternative with different universe and implementation characteristics. Equal target borrowing does not remove those differences. The comparator control excluding external transaction charges also remains below the primary over the full sample; embedded product costs remain in that control. This comparison therefore does not isolate a pure factor premium. Return and risk metrics: historical.metrics.

4.3 Portfolio at 2x versus the Amundi daily 2x ETF

Table 10. Factor portfolio at 2x and Amundi daily 2x: 30 September 2025-31 August 2026 · Product comparison: backtest_products.product_comparison

Short-overlap investment	Cumulative return	Volatility	Drawdown
factor absolute decoupled 2.00x	36.38%	26.22%	-18.74%
amundi 2x	27.51%	23.79%	-17.50%

For this product comparison, the factor portfolio uses a **2x leverage target** with the same allocation and absolute sleeve bands. Its debt is adjusted through the separate leverage band. It is compared with the **Amundi daily 2x ETF** over their common short window. The table reports cumulative returns. The factor portfolio earns the higher return, accompanied by higher volatility and a deeper maximum drawdown. The main strategy and its MSCI World/Core benchmark comparison continue to use **1.25x**.

The factor portfolio includes modelled ongoing trading and borrowing costs. The Amundi return is taken directly from its USD NAV, which incorporates the product's expenses and leveraged financing economics. **The portfolio's borrowing charges are therefore not applied to Amundi.** The Amundi series excludes additional investor purchase/exit brokerage, taxes and listing spreads. Both paths are normalised to the first common post-entry observation; this removes the modelled portfolio's opening expenses.

Amundi resets exposure daily, while the factor portfolio manages leverage within a band. These mechanics produce different exposure paths despite the same nominal 2x label. The evidence is limited to the observed product overlap and does not establish long-period or future superiority. The 2x experiment remains an additional comparison; it leaves the main investment and capital-inflow specifications unchanged.

5. Robustness and Implementation Risk

The growth ranking is not stable across all dates. In independently restarted early and later periods, Core exceeds the main portfolio's CAGR in the early segment, while the main portfolio leads in the later segment. Appendix B retains this counter-evidence and the alpha estimates. Overlapping rolling windows and alternative entry dates do not provide independent probabilities of future success.

The main portfolio's exploratory alpha intervals include zero at both exposure levels. The slightly negative leveraged alpha and the positive matched-target CAGR gap answer different questions. Alpha is a beta-adjusted excess-return intercept against **unlevered Core**; the CAGR gap compares compounded account growth with Core at the **same 1.25x target**. The conditional return-pair bootstrap below addresses uncertainty in that CAGR gap under fixed realised strategy returns; it is separate from the alpha regression. Historical testing, changes in period

rankings and a market-exposed set of equity sleeves prevent a claim of established future superiority. The later restart is retrospective rather than a pristine holdout.

5.1 Paired specification sensitivity and sampling uncertainty

Table 11. Factor minus Core CAGR under the frozen paired variants (percentage points per year) · Paired accounts and sensitivity: paired_robustness.run

Endpoint	Baseline	Median	Minimum	Maximum	Positive
2023-12-29	-0.14	-0.14	-0.34	-0.06	0/14
2024-12-31	-0.24	-0.24	-0.47	-0.18	0/14
2025-12-31	-0.01	-0.01	-0.29	+0.05	5/14
2026-08-28	+0.42	+0.42	+0.17	+0.47	14/14

The original case is reproduced exactly before the variants. The 14 available specifications include the baseline and one-at-a-time changes to the start (one to five common NAV dates later), financing markup (250/350 bp), fund fee (2.5/5 bp), leverage band (0.09/0.11) and sleeve band (4.5/5.5 percentage points). Each pair applies the same change to factor and Core accounts. The five requested earlier starts are unavailable in the admitted data; they are reported as missing, without extrapolation. Core has no factor sleeve bands. Both accounts retain the original first-NAV execution rule. The second-NAV experiment is excluded.

All 14 pairs have a positive CAGR gap at the August 2026 endpoint, while all are negative at the 2023 and 2024 endpoints. Five are positive and nine negative at the end of 2025. The minimum terminal gap occurs with the added 5 bp fee: fees change both accounts' trading paths. The full matrix, event counts and family summaries are retained in the calculation materials. Median and range describe these fixed cases; they are neither a probability distribution nor a significance test. The protocol was frozen before these runs, after earlier historical evidence was known, and is not an untouched preregistration.

Table 12. Conditional stationary block bootstrap of the matched-target CAGR gap · Paired return intervals: paired_robustness.bootstrap

Mean block	Draws	Observed gap (pp)	95% low (pp)	95% high (pp)
20	5,000	+0.42	-0.73	+1.64
60	5,000	+0.42	-0.66	+1.74

Annualised tracking error is **1.97%** and the descriptive information ratio is **0.160**, using the report's nominal-252 convention. For each block length, 5,000 paired circular stationary resamples retain the contemporaneous factor/Core return pair, with fixed seeds and opening charges counted once. Both percentile 95% intervals include zero. Under this conditional resampling model the observed CAGR advantage is not statistically separated from zero. This is not a rule-level resimulation: holdings, financing and threshold decisions are already embodied in the realised returns. The intervals do not remove retrospective selection, prove market efficiency or forecast future performance.

For these no-client fee sensitivities only, monthly fees use the original no-client fund ledger and a final calendar-day stub after the last actual monthly payment, without a duplicate fee at a regular anniversary. This explicit terminal accrual convention does not change the historical 120-month inflow experiment or the zero-fee primary backtest.

5.2 Financing and implementation

Financing and custody costs can change signals, holdings and later trades. Consequently, small changes in charges need not produce monotone ending values. Appendix B reruns the whole strategy at each financing margin, including newly triggered trades. At 0 bp the path has five post-entry interventions, compared with two at 300 bp, including additional leverage adjustments around the 2020 decline. The 13.64% versus 13.24% CAGR comparison therefore includes changed exposure paths as well as direct interest expense. Zero markup still incurs the reference rate. A constant-debt estimate of financing drag cannot reproduce this comparison, and the table does not identify a universally safe borrowing margin. Higher ending wealth in an isolated fee scenario is not evidence that paying higher fees is beneficial.

A larger pool of capital, including possible external subscriptions, might support negotiation of better financing terms. The funding sensitivity measures alternative assumed borrowing margins; it does not establish a relationship between assets and lender pricing. The baseline margin remains fixed, and the companion inflow model does not automatically reduce it as assets grow.

Hypothetical instantaneous losses at maintenance boundaries include both sale-funded cures and insolvency before a cure is possible. These are imposed account states with liquidation friction, not actual historical lender accounts. Positive equity and successful modelled sales do not establish that a lender permits the cure or executes at that NAV.

Actual collateral eligibility, account category, rate fixing, settlement, liquidation timing and complete charges remain unconfirmed. The public banking material supports scenario definitions rather than an individual credit offer. Sparse holdings snapshots cannot explain the entire historical return difference or supply a full historical sector panel.

6. Conclusion

Under the stated conventions, the 60/15/10/15 portfolio with absolute ±5-percentage-point bands and separate leverage management has a favourable full-sample growth/risk comparison with MSCI World/Core. The comparison is specific to the August 2026 endpoint: the accounts were nearly level at the end of 2025, when the CAGR advantage was absent, and the terminal advantage developed in the observed 2026 segment. The simpler common band consciously permits factor drift. Borrowing adds historical growth and materially increases total risk. The unlevered factor mix has a higher historical Sharpe ratio than unlevered Core (0.702 versus 0.672); financing at the modelled 300 bp markup brings the main levered portfolio to 0.665. The paired sensitivities retain a positive terminal gap but reverse the ranking at earlier endpoints; both conditional bootstrap intervals include zero.

Shared access to the factor/leverage implementation is the central investment rationale; the favourable historical comparison is supporting evidence. Any financing advantage from pooling capital remains conditional on applicable lending terms.

The evidence supports further investigation of a fixed specification rather than a forecast of superior returns. Prospective evaluation and verified lending and execution terms are separate requirements from reproducing the historical accounts. The capital-inflow companion report applies the portfolio to a shorter historical fund experiment with hypothetical fees, flows and business costs. Its separate flat, fixed-growth and imposed-loss controls are not projections of the full-period backtest CAGR.

Sources and Data

Fund identities: Core IE00B4L5Y983; Momentum IE00BP3QZ825; Quality IE00BP3QZ601; Value IE00BP3QZB59. Dimensional IE00B2PC0153; Amundi FR0014010HV4.

Market data and financing series

NAV observations and reference rates enter the calculations. Their sample dates are specified in the analysis. Provider links identify the funds and series; the supplied local files are the calculation inputs.

Source	Publisher	Use in this report
Core MSCI World NAV	BlackRock / iShares	Core accumulating USD NAV; investment returns
Momentum NAV	BlackRock / iShares	Momentum accumulating USD NAV; investment returns
Quality NAV	BlackRock / iShares	Quality accumulating USD NAV; investment returns
Value NAV	BlackRock / iShares	Value accumulating USD NAV; investment returns
Indicative reference rate	Federal Reserve Bank of New York	Early indicative financing reference series
Official SOFR	Federal Reserve Bank of New York	Later official financing reference series
Dimensional NAV hardcopy	Bloomberg, author-supplied hardcopy	USD fund NAV; secondary fund comparison
Amundi daily 2x NAV	Amundi	USD ETF NAV; short-window daily 2x product comparison

Cost, product and methodology references

These sources support the stated assumptions, product descriptions and checks. They are distinguished from time-series inputs.

Source	Publisher	Use in this report
Commission tariff	Swissquote	Current CHF commission schedule for the investment cost model
Investor-side stamp duty	Swiss Federal Tax Administration	Basis for the modelled 0.15% investor-side duty
Commission FX conversion	ECB / Official Journal	31 August 2026 reference cross-rate; CHF commission conversion only
Dimensional key information document	Dimensional Funds plc / FundAssist	USD share-class identity and permitted investment universe; 17 July 2025 KID
Reference-rate methodology	Federal Reserve Bank of New York	Interpretation of the early reference-rate proxy
Risk-measure definitions	CFA Institute	Background on risk-adjusted performance; calculation conventions stated in Method
Collateralised loan terms	Swissquote Bank Ltd Switzerland	Contractual context for hypothetical credit and liquidation risks
Lombard loan overview	Swissquote Bank Ltd Switzerland	Product context for borrowing; not an account-specific rate quote
Published Core annual returns; Published Momentum annual returns; Published Quality annual returns; Published Value annual returns	BlackRock / iShares	Rounded calendar-year returns; secondary validation of NAV calculations
Amundi key information document, 28 April 2026	Amundi	Daily-reset 2x mechanism and embedded product/financing costs; not a NAV download source

Original files and download records are retained with the reproducibility materials.

Appendix A. Band Sensitivity

Table A1. Absolute/separate and capped-relative policies · Policy accounts: backtest_workflow.policy_comparison

Rule	Exposure	CAGR	Volatility	Drawdown
Absolute / separate leverage	1.000	11.87%	14.96%	-33.56%
Capped-relative comparison	1.000	11.79%	14.98%	-33.72%
Absolute / separate leverage	1.250	13.24%	18.59%	-39.70%
Capped-relative comparison	1.250	12.92%	18.47%	-39.89%

The capped-relative rule has narrower factor bands, particularly for Quality. Broader bands do not guarantee lower overall risk or fewer trades on every path. Allocation grids and legacy coupled-absolute search maxima are retrospective sensitivity evidence and do not define the separated primary or an optimal future allocation.

Appendix B. Statistical and Period Checks

Table B1. Main portfolio alpha against unlevered Core · Alpha and HAC intervals: historical.metrics

Exposure	Beta	Alpha / year	HAC low	HAC high
1.000	0.980	0.49%	-0.35%	1.33%
1.250	1.219	-0.08%	-1.11%	0.95%

Intervals are exploratory nominal-252 annualised regression-intercept intervals with five-lag HAC errors. They are not adjusted for specification selection or multiple testing. The leveraged regression remains against unlevered Core, so its intercept must not be confused with the matched-target CAGR gap.

Table B2. Independently restarted retrospective periods at 1.25x · Policy accounts: backtest_workflow.policy_comparison

Retrospective period	Investment	Exposure	CAGR	Volatility	Drawdown
Early (retrospective)	Factor portfolio	1.250	13.05%	18.61%	-39.70%
Early (retrospective)	MSCI World Core	1.250	13.15%	18.91%	-40.36%
Later (retrospective)	Factor portfolio	1.250	12.61%	18.20%	-32.94%
Later (retrospective)	MSCI World Core	1.250	11.44%	18.49%	-33.09%

The early period is 3 October 2014-31 December 2021; the later period starts on the first common 2022 observation, 4 January, and ends on 28 August 2026. Each account restarts capital and opening charges. These splits preserve counter-evidence rather than provide an untouched holdout.

Table B3. Financing-markup sensitivity above the reference rate · Borrowing scenarios: backtest_funding.funding_comparison

Markup above reference, bp	Primary full CAGR	Primary later CAGR
0	13.64%	13.30%
300	13.24%	12.61%
500	12.77%	12.13%
1000	11.41%	11.11%

The later result is a restarted account, not a continuation of the full-history balance. Policy discontinuities prevent interpreting a finite scan as a continuous or globally complete break-even search.

Appendix C. Cost Scope and Measurement Conventions

Financing day count. ACT/360 counts actual elapsed calendar days over a 360-day denominator and capitalises interest at NAV dates. It is the model convention used throughout this historical scenario. Clause 4.2 of the saved Swissquote collateralised loan agreement states a 30/360 basis and permits another method indicated on the website. The model therefore does not reconstruct contractual invoices or establish the terms of an individual account.

Calendar and annualisation. The four ETF NAV series and Dimensional share one observed calendar for comparable account measurements. Missing investment observations are excluded, not interpolated. The nominal 252 convention is applied consistently to volatility, excess-return Sharpe and regression annualisation; it is not the measured number of common observations per calendar year. CAGR uses elapsed calendar years. These conventions remain unchanged.

Custody sensitivity. Custody charges are excluded from the main account. Separate whole-path scenarios retain private-account and legal-entity tariff assumptions, including VAT, without selecting a legal form or asserting negotiated terms. The common reconstruction produces custody/custody_metrics.csv in the custody job; the complete backtest evidence records Table backtest_22. See the custody scenario configuration and custody account simulation. Charging custody can change debt, band crossings and subsequent trades; it is not a fixed deduction from CAGR.

Boundary display. The minimum in Table 8 rounds to 1.150x but is slightly below the lower boundary. It triggers the next-NAV adjustment on 9 April 2021.

Supplementary index label. The supplementary Bloomberg input **MXWOLDNU Index** is treated as a daily 2x leveraged index in the product diagnostics. Its original filename and source identifier are preserved. It is separate from the Core ETF used as the main benchmark and from the Amundi ETF NAV used in Table 10.

Reproducibility Materials

The blue links open the responsible function or configuration on GitHub. The calculation map identifies exact rows, fields and generated evidence paths. Code links use fixed snapshots. The prepared code and separately supplied original data reconstruct the accounts; full analyses and source-cell bindings remain in the accompanying calculation materials. Provider originals stay local.

AI Assistance and Responsibility

Tools, scope and author contribution

OpenAI Codex (OpenAI, n.d.) assisted method development, programming, data processing, portfolio and fund simulations, statistics, interpretation and documentation. AI-generated or substantially revised wording appears throughout both abstracts, main texts, source commentary and appendices. Tables and figures were produced through AI-assisted code. Alex supplied the materials, specified and approved principal design choices, selected comparisons and reviewed drafts; the author retains responsibility for the content.

During the October 2026 revision, Alex also supplied a **Claude (Anthropic, n.d.)** review of the reports. Its methodological, wording and layout suggestions were evaluated against project evidence and informed this revision. Claude was not the original calculation engine. Its model/version has not been established.

List of aids (Hilfsmittelverzeichnis)

Tool	Purpose and affected material
OpenAI Codex	Methodology, code, data processing, calculations, statistics, interpretation, wording throughout both reports, tables, figures and documentation
Claude	Author-supplied critical review informing the revised interpretation, wording and presentation of both reports
Python, NumPy, pandas, openpyxl and Matplotlib	Calculation implementation, spreadsheet inputs, tables and charts
pytest	Automated calculation and report checks
pypdf, pdfplumber, ReportLab, Pillow, PDFium and Poppler	Source-PDF extraction, report generation and text, link and visual checks

Verification and retained records

Checks covered accounting identities, reference comparisons, separate implementations, numerical bindings, text and layout. They assess implementation under the stated inputs; they do not constitute independent human expert review. Provider sources retain their attribution.

The AI-use record documents assistance and verification limits. Selected prompts, passages from the supplied Claude review and working records are retained privately. No complete chronological prompt/output or historical model-version archive has been established. References identify applications; assessment-specific documentation remains the author's responsibility.

Application references

OpenAI. (n.d.). *Codex* [Generative AI coding assistant]. Application reference.

Anthropic. (n.d.). *Claude* [Generative AI assistant; author-supplied review]. Application reference.

This declaration concerns Alex's research contribution. It does not attribute unverified AI use to other group members or future video narration.